

B.TECH. COMPUTER SCIENCE AND ENGINEERING (AI& ML)
Deenbandhu Chhotu Ram University of Science & Technology Murthal, Sonapat, Haryana
Semester – III (Effective from Session 2025-26)

Course Code: PCCSE207	Minor	TCA	TES	PCA	PES	Total
Title of the Course: COMPUTER ORGANIZATION AND ARCHITECTURE	15	25	60	-	-	100
L-T-P: 3-1-0						
Credits: 4	Duration of Exam: 3 Hours					
Brief Write-Up About the Course(100-150 words) This course offers a comprehensive overview of the fundamental structure, functionality, and design of digital computers. It begins with the block-level understanding of digital computers and explores concepts of register transfer, micro-operations, and instruction cycle execution. Students learn the distinction between CISC and RISC architectures and examine memory-reference instructions, input-output systems, and interrupt handling. The course further delves into data representation techniques and computer arithmetic, including algorithms for fixed and floating-point operations. Advanced topics such as microprogrammed control, CPU architecture, addressing modes, and I/O organization are also covered. Finally, the course explores memory hierarchy, cache, associative memory, pipelining techniques, vector processing, and multiprocessor systems with interconnection and synchronization mechanisms. This subject builds a strong foundation for understanding how computers operate at the hardware level and prepares students for advanced computer architecture and systems programming.						
Course Objective(CO): 1. The purpose of the course is to introduce principles of computer organization and the basic architectural concepts. 2. To introduce data types and representation of computer arithmetic. 3. To study microprogrammed control unit, addressing modes and data transfer techniques. 4. To study memory organization, pipelining and vector processing.						
UNIT - I Digital Computers: Introduction, Block diagram of Digital Computer, Definition of Computer Organization, Computer Design and Computer Architecture. Register Transfer Language and Micro operations: Register Transfer language, Register Transfer, Bus and memory transfers, Arithmetic Micro operations, logic micro- operations, shift micro- operations, Arithmetic logic shift unit. Basic Computer Organization and Design: Instruction codes, Computer Registers Computer instructions, Timing and Control, Instruction cycle, Memory Reference Instructions, Input – Output and Interrupt. Reduced Instruction Set Computer: CISC Characteristics, RISC Characteristics. UNIT - II Data Representation: Data types, Complements, Fixed Point Representation, Floating Point Representation. Computer Arithmetic: Addition and subtraction, multiplication Algorithms, Division Algorithms, Floating – point Arithmetic operations. Decimal Arithmetic unit, Decimal Arithmetic operations. UNIT - III Microprogrammed Control: Control memory, Address sequencing, micro program example, design of control unit. Central Processing Unit: General Register Organization, Instruction Formats, Addressing modes, Data Transfer and Manipulation, Program Control.						

Input-Output Organization: Input-Output Interface, Asynchronous data transfer, Modes of Transfer, Priority Interrupt Direct memory Access.

UNIT IV

Memory Organization: Memory Hierarchy, Main Memory, Auxiliary memory, Associate Memory, Cache Memory. Pipeline and Vector Processing: Parallel Processing, Pipelining, Arithmetic Pipeline, Instruction Pipeline, RISC Pipeline, Vector Processing, Array Processor.

Multi Processors: Characteristics of Multiprocessors, Interconnection Structures, Inter processor arbitration, Inter processor communication and synchronization, Cache Coherence.

Text book:

1. Computer System Architecture – M. Moris Mano, Third Edition, Pearson/PHI.

Reference books:

1. Computer Organization – Car Hamacher, ZvonksVranesic, SafeaZaky, Vth Edition, McGrawHill.
2. Computer Organization and Architecture – William Stallings Sixth Edition, Pearson/PHI.
3. Structured Computer Organization – Andrew S. Tanenbaum, 4th Edition, PHI/Pearson.

CourseOutcome(CO):

After successful completion of the course, students will be able:

1. To understand principles of computer organization and the basic architectural concepts.
2. To understand various data types and their representation.
3. To understand microprogrammed control unit, addressing modes and data transfer techniques.
4. To understand various techniques of memory organization, pipelining and vector processing.